

GEOTECHNICAL INVESTIGATION REPORT

Riverfront Plaza Development

42 Riverfront Avenue

Project Reference: PROJ-003

Date: March 2026

1. Executive Summary

This geotechnical investigation was carried out for the proposed Riverfront Plaza development at 42 Riverfront Avenue. Three boreholes (BH-1, BH-2, BH-3) were advanced to depths of 15 m. The site is underlain by approximately 6 m of loose fill material overlying dense gravels. The fill layer presents challenges for shallow foundations and should be considered in all structural design decisions.

2. Soil Profile and Stratigraphy

Borehole logs indicate the following stratigraphy from surface:

0–6 m: Loose fill comprising demolition rubble, sand, and clay. Standard Penetration Test (SPT) N-values range from 3 to 8, indicating very loose to loose consistency.

6–15 m: Dense sandy gravel with cobbles. SPT N-values range from 22 to 35, indicating dense to very dense material. This stratum provides suitable bearing for foundations.

3. Foundation Recommendations

Based on the soil conditions encountered, the following foundation options are recommended:

Shallow Foundations: Strip or pad footings founded at a minimum depth of 2.5 m below finished floor level, bearing on the dense gravel stratum. The allowable bearing capacity at this level is 120 kPa. Settlement is estimated at less than 25 mm.

Driven Piles: As an alternative for heavily loaded columns, driven piles should be taken to a minimum depth of 14 m to develop adequate skin friction and end bearing in the dense gravel. This option is recommended where column loads exceed 500 kN.

4. Groundwater Conditions

Groundwater was encountered during drilling at a depth of 1.8 m below existing ground level. Seasonal variation of ± 0.5 m is anticipated based on regional data. Temporary dewatering will be required during construction of any excavations below 1.5 m depth. Permanent waterproofing measures should be provided for all below-ground structures. Groundwater is classified as mildly aggressive to concrete (Class XA1 per EN 206).

5. Seismic Assessment and Liquefaction

The site is located in a moderate seismic zone with a design peak ground acceleration of $I_a = 0.18g$. The loose saturated sands within the fill layer between 1 m and 4 m depth are susceptible to liquefaction during a design earthquake event. Seismic amplification effects are expected due to the soft fill overlying dense gravel. Mitigation measures including vibro-compaction or removal and replacement of the fill are recommended prior to construction. Dynamic compaction is also a viable alternative.